

Necessary and Sufficient Multiplicity Conditions for Noether's Fundamental Theorem of Algebraic Functions

By H. Kapferer. With an addendum, jointly with E. Noether. Original publication: Math. Ann. 97 (1927), pp. 559–567.

What follows is a sharpening of Noether's fundamental theorem, in the form of necessary and sufficient multiplicity conditions. I formulate the facts comprised by Noether's fundamental theorem - following the ideal-theoretic conception of the theorem, but without using ideal-theoretic concepts and results either here or below - as the following two separate assertions.

Theorem Ia. For each common zero

$$x = \alpha, \quad y = \beta$$

of two relatively prime polynomials φ, ψ in x, y , with arbitrary complex coefficients, there exists a smallest positive integer ρ such that the module

$$(\varphi, \psi, \mathfrak{p}^\rho)$$

denotes exactly the same set of polynomials as the module

$$(\varphi, \psi, \mathfrak{p}^{\rho+u}),$$

where

$$\mathfrak{p} = (x - \alpha, y - \beta)$$

and ν is any positive integer.

Theorem Ib. Denote this uniquely determined set of polynomials by $\mathfrak{q}$, and by $\mathfrak{q}_i$ when the

zero in question is the i -th of the s distinct common zeros,

$$x = \alpha_i, \quad y = \beta_i.$$

Then a necessary and sufficient condition that a prescribed polynomial K in x, y have the divisibility property

$$K = 0(\varphi, \psi)$$

is the simultaneous fulfillment of the s congruences

$$K = 0(\mathfrak{q}_1), \quad K = 0(\mathfrak{q}_2), \quad \dots, \quad K = 0(\mathfrak{q}_s).$$

The notation is chosen to recall the agreement in meaning with the notions now called the prime ideal $\mathfrak{p}$ and the primary ideal $\mathfrak{q}$. Historically, as is clear from the introduction to Max Noether's basic paper on the subject, the fundamental theorem grew out of the attempt to solve the following fundamental problem. Given any polynomial K in x, y , one wants to decide whether it is divisible by (φ, ψ) ; equivalently, whether there exist two further polynomials A and μ in x, y such that

$$K = A\varphi + \mu\psi$$

is an identity in x, y .

Noether's theorem does not itself solve this problem. It is rather a fundamental reduction of the problem to a simpler one: the question of necessary and sufficient conditions for the individual congruences

$$K = 0(\mathfrak{q})$$

to hold. Only when this latter question has been answered is Max Noether's original problem settled. This is the aim of the present paper. It is reached by giving finitely many multiplicity conditions, easily checked in practice, which are necessary and sufficient for $K = 0(\mathfrak{q})$.

A sufficient condition for $K = 0(\mathfrak{q})$, though not a necessary one, is already known and is often used and cited. It is usually put as follows: for the congruence $K = 0(\varphi, \psi)$ it is enough that every intersection point of

$$\varphi = 0, \quad \psi = 0$$

occur in the polynomial K with sufficiently high multiplicity. This statement is contained in Theorem I as a special case. An ρ -fold point of K is the same as $K = 0(\mathfrak{p}^\rho)$, hence a fortiori

$$K = 0(\varphi, \psi, \mathfrak{p}^\rho),$$

that is, $K = 0(\mathfrak{q})$. But the converse is not true.

The promised solution can be described briefly. To the individual $\mathfrak{q}$ one can assign a system of t positive integers

$$(g_1), (g_2), \dots, (g_t),$$

and to the module $(K, \mathfrak{q})$ the same number of positive or negative integers

$$(K_1), (K_2), \dots, (K_t),$$

so that the following main theorem holds.

Theorem II. A necessary and sufficient condition for

$$K = 0(\mathfrak{q})$$

is the simultaneous fulfillment of the t conditions

$$(K_1) \geq (g_1), \quad (K_2) \geq (g_2), \quad \dots, \quad (K_t) \geq (g_t).$$

Moreover

$$(g_1) + (g_2) + \dots + (g_t) = \mu,$$

where μ denotes the intersection multiplicity attached to $\mathfrak{q}$, as defined by the resultant.

For the proof we may suppose, without loss of generality, that the zero belonging to $\mathfrak{q}$ is

$$x = 0, \quad y = 0.$$

If $A(x, y)$ is any polynomial in x, y , the symbol (A) will denote the exponent with which y occurs as a factor of $A(0, y)$. If $A(0, y)$ vanishes identically, we put

$$(A) = \infty.$$

The numbers $(g_1), (g_2), \dots, (g_t)$ assigned to $\mathfrak{q}$ are defined by the following system of equations, which is the methodological basis of the procedure:

$$\begin{aligned} f_1 \frac{g_1(0, y)}{y^{(g_1)}} - g_1 \frac{f_1(0, y)}{y^{(g_1)}} &= xh_1, \quad f_2 \frac{g_2(0, y)}{y^{(g_2)}} - g_2 \frac{f_2(0, y)}{y^{(g_2)}} = xh_2, \quad \vdots \\ f_r \frac{g_r(0, y)}{y^{(g_r)}} - g_r \frac{f_r(0, y)}{y^{(g_r)}} &= xh_r. \end{aligned} \tag{1}$$

This system is constructed solely from the given pair of polynomials φ, ψ . The polynomials f_1, g_1 are defined to be identical with φ, ψ , possibly after interchanging their order, in such a way that

$$(f_1) \geq (g_1).$$

With this choice, and only with this choice, h_1 is a polynomial. The polynomials f_2, g_2 are then taken to be h_1, g_1 , again ordered so that

$$(f_2) \geq (g_2).$$

In general the pair f_r, g_r is the pair h_{r-1}, g_{r-1} , ordered by the side condition

$$(f_r) \geq (g_r).$$

The system (1) may be continued indefinitely in this manner. The main theorem, however, requires only the first t equations, where t is determined by the following fact.

Lemma I. There exists a natural number t , in fact $t \leq \mu$, where μ is the intersection multiplicity with which the point $x = 0, y = 0$ occurs in $\varphi = 0, \psi = 0$, such that the first t of the numbers defined by (1), namely

$$(g_1), (g_2), \dots, (g_t),$$

are all greater than zero, while (g_{t+1}) and all following numbers are zero.

Indeed, using the product theorem for resultants, the intersection multiplicity at $(0, 0)$ of

$$f_r = 0, \quad g_r = 0$$

is

$$\mu - (g_1) - (g_2) - \dots - (g_r).$$

This proves that the process stops after at most μ steps. It also gives

$$\mu = (g_1) + (g_2) + \dots + (g_t), \tag{2}$$

a formula which is noteworthy because it gives, in complete generality, a rational and practically easy method for determining the intersection multiplicity once the point in question is known.

After the numbers (g_i) have been found, the numbers (K_i) are determined by the following system, corresponding to (1):

$$\begin{aligned} K_1 \frac{g_1(0, y)}{y^{(g_1)}} - g_1 \frac{K_1(0, y)}{y^{(g_1)}} &= x K_2, \quad K_2 \frac{g_2(0, y)}{y^{(g_2)}} - g_2 \frac{K_2(0, y)}{y^{(g_2)}} = x K_3, \quad \vdots \\ K_r \frac{g_r(0, y)}{y^{(g_r)}} - g_r \frac{K_r(0, y)}{y^{(g_r)}} &= x K_{r+1}. \end{aligned} \tag{3}$$

We put $K_1 = K$. It is clear that K_2 is a polynomial if and only if

$$(K_1) \geq (g_1).$$

If this condition is satisfied, then K_3 is a polynomial if and only if

$$(K_2) \geq (g_2).$$

In general, provided that K_r is already a polynomial, K_{r+1} is a polynomial if and only if

$$(K_r) \geq (g_r).$$

The proof of the main theorem rests on the following further lemma.

Lemma II. Put

$$\mathfrak{q}^{(i)} = (f_i, g_i, \mathfrak{p}^\rho).$$

Then for every $i = 1, 2, \dots$, the congruence

$$K_i = 0(\mathfrak{q}^{(i)})$$

holds if and only if both of the following conditions hold simultaneously:

$$(K_i) \geq (g_i), \quad K_{i+1} = 0(\mathfrak{q}^{(i+1)}).$$

The proof of the lemma is obtained by combining the systems (1) and (3). Since, by Lemma I, g_{t+1} no longer vanishes at the origin, the module $\mathfrak{q}^{(t+1)}$ consists of all polynomials; in other words it is the unit ideal. Thus

$$K_{t+1} = 0(\mathfrak{q}^{(t+1)})$$

is automatic. Applying Lemma II t times gives the proof of the main theorem.

Two additions to the main theorem are useful. Their proofs are given in my longer treatment of the same subject.

Addition I. The number t occurring in the main theorem is identical with the smallest positive integer v such that

$$x^v = 0(\mathfrak{q}).$$

Addition II. The systems of equations (1) and (3), together with the systems obtained from them by interchanging x and y , give, after suitable combination, a rational method for the exact numerical determination of the exponent ρ belonging to $\mathfrak{q}$, that is, of the minimal exponent ρ in Theorem Ia.

Bertini gave a general formula for ρ . But that formula does not always give the minimal value of ρ ; this is shown in the cited paper by the example

$$\varphi = x^3 + y^4, \quad \psi = x^2 - y^3.$$

The main theorem also supplies a new proof of Bertini's formula.

An interesting specialization of the theorem follows from the restrictive assumption that the zero belonging to

$$\mathfrak{q} = (\varphi, \psi, \mathfrak{p}^\rho)$$

is a simple point on at least one of the two curves $\varphi = 0$, $\psi = 0$. If it is, say, a simple point on ψ , then the number t in the theorem is exactly equal to μ , and the t inequalities of the theorem may be replaced by a single multiplicity condition. This gives the following general statement.

Theorem III. Suppose that all intersection points of

$$\varphi = 0, \quad \psi = 0$$

are simple points of ψ . Assume also that K is relatively prime to ψ . Then a necessary and

sufficient condition for the congruence

$$K = 0(\varphi, \psi)$$

is that K vanish at all these points in such a way that, at each point, the intersection multiplicity in the pair of curves

$$K = 0, \quad \psi = 0$$

is at least as large as in the pair

$$\varphi = 0, \quad \psi = 0.$$

This third theorem is noteworthy for another reason. The multiplicity condition it requires is one whose fulfillment can be checked, independently of formula (2), by carrying out differentiation finitely many times. This follows at once from the following result, which I stated in 1923.

If A and B are any two relatively prime polynomials in x, y , and P is a common point which is a simple point of B , then the multiplicity of P as an intersection point is equal to a number μ if and only if the multiplicity of the same point in the pair of curves

$$A' = 0, \quad B = 0$$

is exactly $\mu - 1$, where

$$A' = \frac{\partial A}{\partial x} \frac{\partial B}{\partial y} - \frac{\partial A}{\partial y} \frac{\partial B}{\partial x}.$$

Finally, the main theorem can, with a small modification, also be transferred to homogeneous polynomials in three variables x, y, z . This transfer is especially elegant in the special case just treated.

Addition to Theorem III. Theorem III remains entirely unchanged, even literally, if

φ, ψ, K are understood to be homogeneous polynomials in three variables.

Addendum, jointly with E. Noether

The system of equations (1) also gives a rational method for determining a complete residue system modulo $\mathfrak{q}$. By this is meant a complete system of representatives for the finitely many residue classes which are linearly independent modulo $\mathfrak{q}$ over the coefficient domain P . The separate steps in the proof of the main theorem thereby become transparent. At the same time one obtains, as a byproduct, the fact that the resultant multiplicity μ agrees with the length of the ideal $\mathfrak{q}$, that is, with the rank of the residue-class ring, or the number of residue classes linearly independent over P .

The complete residue system is characterized as follows, using the notation of (1) and Lemma II.

Theorem IV. A complete residue system modulo $\mathfrak{q}$ is given by the polynomials

$$h_i, h_i y, h_i y^2, \dots, h_i y^{(g_i)-1} \quad (i = 1, 2, \dots, t),$$

taken for all i . More precisely, for each i , the block

$$h_i, h_i y, \dots, h_i y^{(g_i)-1}$$

gives a complete residue system of $\mathfrak{q}^{(i+1)}$ modulo $\mathfrak{q}^{(i)}$.

From the definition one obtains immediately

$$\mathfrak{q}^{(i+1)} = (\mathfrak{q}^{(i)}, h_i).$$

Thus Theorem IV will be proved once the following relations have been shown:

$$h_i x = 0(\mathfrak{q}^{(i)}), \quad h_i y^{(g_i)} = 0(\mathfrak{q}^{(i)}), \quad (4)$$

and

$$h_i F(y) = 0(\mathfrak{q}^{(i)}) \implies F(y) = 0(y^{(g_i)}). \quad (5)$$

The relations (4) assert that the linear combinations of

$$h_i, h_i y, \dots, h_i y^{(g_i)-1}$$

exhaust the residue classes of $\mathfrak{q}^{(i+1)}$ modulo $\mathfrak{q}^{(i)}$, while (5) expresses their linear independence. Since $\mathfrak{q}^{(t+1)}$ is the unit ideal - a fact now following from the finite length of $\mathfrak{q}$ - this proves Theorem IV. It also proves that the length of $\mathfrak{q}$ is

$$(g_1) + (g_2) + \dots + (g_t),$$

and hence, by (2), agrees with the resultant multiplicity.

The first relation in (4) is read directly from (1). The second follows as follows. We have

$$h_i g_i = 0(\mathfrak{q}^{(i)}).$$

Using the first relation in (4), this gives

$$h_i g_i(0, y) = 0(\mathfrak{q}^{(i)}),$$

or

$$h_i y^{(g_i)} \frac{g_i(0, y)}{y^{(g_i)}} = 0(\mathfrak{q}^{(i)}).$$

Since

$$\frac{g_i(0, y)}{y^{(g_i)}} \not\equiv 0 \pmod{(y)},$$

it follows that

$$h_i y^{(g_i)} = 0(\mathfrak{q}^{(i)}).$$

For the proof of (5), note first the immediate consequence of (1): the polynomials f_i and g_i can have as a common divisor only a polynomial $d(y)$ in y which is not divisible by y . Thus

$$f_i = d_i(y)\bar{f}_i, \quad g_i = d_i(y)\bar{g}_i,$$

where $\bar{f}_i$ and $\bar{g}_i$ are relatively prime. Since $d_i(y) \not\equiv 0 \pmod{\mathfrak{p}}$, the ideal $\mathfrak{q}^{(i)}$ is also

$$(\bar{f}_i, \bar{g}_i, \mathfrak{p}^\rho).$$

For every sufficiently large ρ , this is equal to $(\bar{f}_i, \bar{g}_i, \mathfrak{p}^\rho)$ in such a way that $\mathfrak{q}^{(i)}$ is the primary component belonging to the origin in the ideal $(\bar{f}_i, \bar{g}_i)$.

Let the zeros of $(\bar{f}_i, \bar{g}_i)$ distinct from $(0, 0)$ be

$$(\gamma_1, \delta_1), \dots, (\gamma_r, \delta_r),$$

and let $\tau_1, \dots, \tau_r$ be the exponents of the corresponding primary components of $(\bar{f}_i, \bar{g}_i)$.

Choose the indeterminate u in the product

$$\prod_{j=1}^r \left((x - \gamma_j) + u(y - \delta_j) \right)^{\tau_j}$$

so that after specialization the product remains non-zero modulo $\mathfrak{p}$. If $G(x, y)$ is the product of this specialized polynomial with $d_i(y)$, then

$$G(x, y) \not\equiv 0 \pmod{\mathfrak{p}}, \quad \mathfrak{q}^{(i)}G(x, y) = 0(f_i, g_i).$$

From the assumption

$$h_i F(y) = 0(\mathfrak{q}^{(i)})$$

it follows that

$$h_i F(y)G(x, y) = 0(f_i, g_i).$$

Together with (1), this may be written in the form of identities, and it yields

$$f_i(xA - F(y)H) = 0(g_i), \quad H = G(x, y) \frac{g_i(0, y)}{y^{(g_i)}} = 0(\mathfrak{p}).$$

Since f_i and g_i are relatively prime after removal of the factor $d_i(y)$, one obtains

$$xA - F(y)H = 0(g_i).$$

Hence

$$F(y)H = 0(g_i, x) = (y^{(g_i)}, x),$$

and therefore

$$F(y) = 0(y^{(g_i)}).$$

This proves (5), and hence Theorem IV.

Remark I. Relations (4) and (5), together with Lemma II, may be written as reciprocal relations between ideal quotients:

$$\mathfrak{q}^{(i)} : \mathfrak{q}^{(i+1)} = (\mathfrak{q}^{(i)}, x) = (y^{(g_i)}, x), \quad \mathfrak{q}^{(i)} : (\mathfrak{q}^{(i)}, x) = \mathfrak{q}^{(i+1)}. \quad (6)$$

Here the proof of the second relation also gives a proof of Lemma II. The first relation is the combination of (4) and (5). For the second one, observe that

$$x\mathfrak{q}^{(i+1)} = 0(\mathfrak{q}^{(i)})$$

was proved in (4), and that the converse follows in the same way as (5).

Indeed, from

$$xM = 0(\mathfrak{q}^{(i)})$$

one obtains as above

$$xMG = 0(f_i, g_i),$$

hence

$$xMG \frac{g_i(0, y)}{y^{(g_i)}} = 0(xh_i, g_i).$$

Because $g_i \not\equiv 0 \pmod{x}$ by definition, this gives

$$M \left(G \frac{g_i(0, y)}{y^{(g_i)}} \right) = 0(h_i, g_i) = 0(\mathfrak{q}^{(i+1)}),$$

and therefore

$$M = 0(\mathfrak{q}^{(i+1)}).$$

The necessity in Lemma II follows as follows. If

$$K_i = 0(\mathfrak{q}_i),$$

then

$$K_i(0, y) = 0(y^{(g_i)}, x),$$

so that $(K_i) \geq (g_i)$. By (3) this gives

$$xK_{i+1} = 0(\mathfrak{q}^{(i)}),$$

and hence, by the second relation in (6),

$$K_{i+1} = 0(\mathfrak{q}^{(i+1)}).$$

Sufficiency is read off immediately. Applying Lemma II finitely many times gives the main theorem. Iterating the second relation in (6) also gives Addition I: namely

$$\mathfrak{q} : x^t = \mathfrak{q}^{(t+1)},$$

which is the unit ideal, and t is the lowest power of this kind because

$$\mathfrak{q} : x^{t-1} = \mathfrak{q}^{(t)}.$$

Remark II. If K is not divisible by $\mathfrak{q}$, one can say more. If

$$K = 0(\mathfrak{q}^{(i+1)}), \quad K \neq 0(\mathfrak{q}^{(i)}),$$

then

$$(K, \mathfrak{q}^{(i)}) = (h_i y^\lambda, \mathfrak{q}^{(i)}).$$

For by Theorem IV one has, modulo $\mathfrak{q}^{(i)}$,

$$K = h_i y^\lambda c(y), \quad c(y) \not\equiv 0 \pmod{(y)}.$$

Hence

$$(K, \mathfrak{q}^{(i)}) = (h_i y^\lambda, \mathfrak{q}^{(i)}).$$

Moreover

$$y^{(g_i)-\lambda} K = 0(\mathfrak{q}^{(i)}),$$

and by (5) this is the lowest power with that property. By Lemma II, however, $(g_i) - (K)$ is this lowest power; hence $\lambda = (K)$.

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